Similarity-transformed Hamiltonian and ECC diagrams

Signatures are (created particles, created holes, annihilated particles, annihilated holes), all relative to the reference. Open operator legs are short stubs: creation upward, annihilation downward. All arrows are shown in the operator/ECC layout. Crossings without a dot are not contractions.

Bare Hamiltonian: all thirteen vertices

Vertex 0

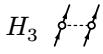
Signature: (1, 0, 1, 0)

Vertex 1

Signature: (0, 1, 0, 1)

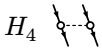
Vertex 2

Signature: (0, 0, 1, 1)



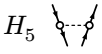
Vertex 3

Signature: (2, 0, 2, 0)



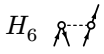
Vertex 4

Signature: (0, 2, 0, 2)



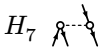
Vertex 5

Signature: (1, 1, 1, 1)



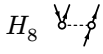
Vertex 6

Signature: (1, 0, 2, 1)



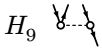
Vertex 7

Signature: (0, 1, 1, 2)



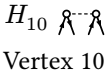
Vertex 8

Signature: (2, 1, 1, 0)



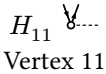
Vertex 9

Signature: (1, 2, 0, 1)



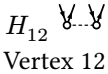
Vertex 10

Signature: (0, 0, 2, 2)



Vertex 11

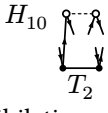
Signature: (1, 1, 0, 0)



Vertex 12

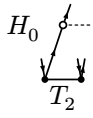
Signature: (2, 2, 0, 0)

Partial contractions and scalar terms



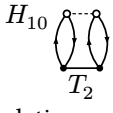
Annihilation sector

Signature: (1, 2, 1, 2)



Unused creation legs

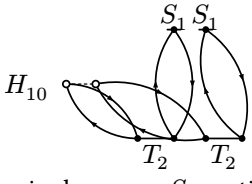
Signature: (2, 2, 0, 0)



Correlation energy

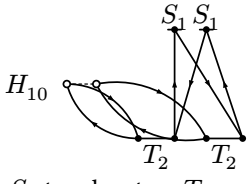
Signature: (0, 0, 0, 0)

ECC: incidence matters



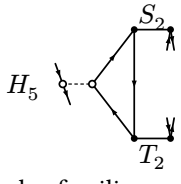
Two single-source S_1 vertices

Signature: (0, 0, 0, 0)



Each S_1 touches two T_2 vertices

Signature: (0, 0, 0, 0)

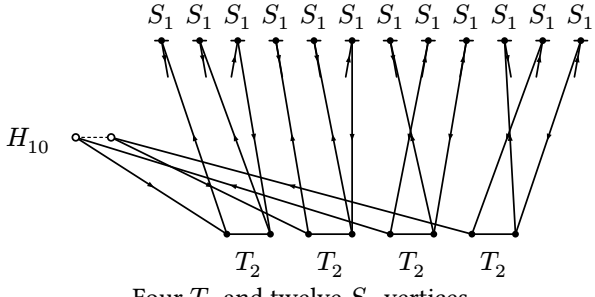


All three edge families; open S legs

Signature: (1, 2, 1, 2)

Both scalar graphs are valid in original ECC coordinates. Only the middle graph meets the stronger doubly-linked energy rule. The renderer does not filter diagrams or change amplitude coordinates.

The S family is not restricted to four vertices



Four T_2 and twelve S_1 vertices

Signature: (0, 0, 6, 6)